A versatile pulsed *X*-band ENDOR spectrometer

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A versatile high power *X*-band (8.5–9.5 GHz) pulsed EPR/ENDOR (electron-nuclear double resonance) spectrometer which can generate hundreds of microwave (MW) and rf pulses is described. The pulse programmer is constructed from a word generator with 32 channels and 4 ns resolution, coupled to five digital delay generators which can produce a total of ten pulses with a resolution better than 1 ns. The spectrometer contains two MW and two rf channels that allow independent variation of the frequency, amplitude, and phase of the MW and rf pulses. The ENDOR probe head is based on a bridged loop gap (BLG) resonator, coupling is achieved *via* a coupling loop connected to a waveguide, and the rf coil serves as a MW shield as well. The adjustment of the coupling is done by an up/down motion of the of the resonator assembly with respect to the fixed coupling loop. A flexible and user friendly data acquisition program written in C++ (Borland version 4.5), which uses the Windows-95 Multiple Document Interface (MDI) programming model, was developed to run the spectrometer. This program allows easy programming of any pulse sequence with sophisticated phase cycling. The performance of the spectrometer is demonstrated by two experiments. The first is the triple resonance hyperfine-selective (HS) ENDOR experiment carried out on a frozen solution of the copper protein laccase. The second is the two-dimensional hyperfine-ENDOR (HYEND) correlation experiment performed on a single crystal of γ-irradiated malonic acid. © *1998 American Institute of Physics.* [S0034-6748(98)02908-6]

I. INTRODUCTION

In the last decade pulsed EPR spectroscopy has undergone considerable development. Sophisticated new pulse schemes have been introduced, and there has been a large increase in the applications of various experiments to a wide range of problems in physics, chemistry, and biology.[1] The new pulse schemes include a variety of pulsed electron-nuclear double resonance (ENDOR) experiments which combine the excitation of both electron and nuclear spins.[2] These experiments require the generation of a number of microwave (MW) and radio frequency (rf) pulses, with variable amplitudes, phases, and frequencies and are often combined with elaborate phase cycling schemes. In the course of an experiment one or more of these parameters are varied, including time intervals, frequency and phase of the MW and rf pulses, strength of the external magnetic field, B_0, or any combination of the above.[1]

Examples of sophisticated experiments based on the nuclear modulation effect are the now standard four-pulse two-dimensional (2D) hyperfine sublevel correlation (HYSCORE) experiment,[3] the five-pulse 2D double nuclear coherence transfer (DONUT) HYSCORE experiment,[4] the six-pulse HYSCORE experiment,[5] dead time and blind spot free four-pulse electron spin echo envelope modulation (ESEEM) experiments,[6] and pulse train detected ESEEM.[7] Examples of sophisticated pulsed ENDOR experiments are the hyperfine-selective ENDOR,[8] requiring microwave pulses with two different frequencies, pulsed triple-ENDOR,[9] where two rf pulses of different frequencies are applied, time-domain ENDOR,[10] remote Mims ENDOR[11] and sublevel-echo ENDOR.[10] Experiments with a higher degree of complexity are the 2D hyperfine correlated ENDOR (HYEND),[12] the chirp ENDOR-HYSCORE,[13] SEDOR-ENDOR,[14] and multiple-quantum ENDOR.[15,16] In addition, there are experiments which require B_0 field jumps such as the 2D field step electron–electron double resonance (FS-ELDOR),[17] and the hole burning experiments, where shaped MW pulses are used.[18]

The design of our original home-built pulsed EPR spectrometer[19] followed the standard design of most operating home-built spectrometers, many of which were constructed more than a decade ago. This spectrometer could not run most of the experiments mentioned above because the number of pulses was limited, the implementation of phase cycling was not straightforward, and the software was not flexible enough to accommodate many experiments. Therefore, in order to keep up with the rapid developments in the field, an upgrade became essential.

In the present report we describe the modifications of our pulsed *X*-band EPR spectrometer that improve its performance. Originally limited to a maximum of four MW pulses (or three MW pulses and one rf pulse), the spectrometer is now a versatile pulse EPR/ENDOR spectrometer that can generate hundreds of MW and rf pulses. The upgrade consisted of: (i) the modification of the MW bridge to include two independent channels with different frequencies, (ii) the design of a new rf unit, (iii) the construction of an ENDOR probe head, (iv) the setup of a new powerful pulse programmer system, and (v) the development of flexible and user friendly software for the data acquisition and pulse programming.

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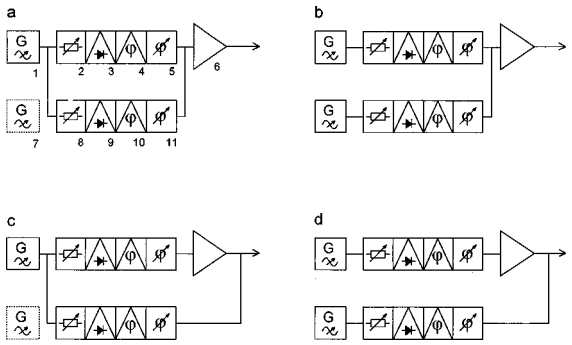


FIG. 1. A schematic representation of the four configurations of the MW channels. The components designated by Nos. 2–5 and 8–11 correspond to channels I and II, respectively. 1,7–MW sources; 6–TWT amplifier; 2,8–manual attenuators; 3,9–MW switches; 4,10–digital phase shifters; 5,11–manual phase shifters. (a) The MW pulses of both channels have high power and the same frequency, but may differ in their amplitude and phase, (b) the same as (a) but with different frequency sources for channels I and II, (c) same as (a) but with a low power output for channel II, (d) same as (b) but with a low power output for channel II.

II. SPECTROMETER DESCRIPTION

The spectrometer contains the following main units: electromagnet, microwave bridge, rf unit, pulse programmer, data acquisition electronics, a number of probe heads, and a computer (Pentium PC, with the Windows 95 operating system). The various devices are interfaced to the computer via a GPIB card (National Instruments), except for the rf synthesizer which is interfaced via a Digital I/O board (National Instruments PC-DIO-96). A data acquisition and control card (DAS1600, Keithley-Metrabyte) is used to digitize analog signals and to trigger the pulse programmer. In the following we give a detailed description of the parts of the spectrometer that have been upgraded.

A. MICROWAVE BRIDGE

The microwave bridge is based on the original design, published earlier.[19] It was modified to accommodate two MW-pulse channels, allowing independent variation of their frequency, amplitude, and relative phase. The microwave sources of the two channels are Gunn diode oscillators (M/A-COM, MA-87840) with a frequency range of 8.5–9.5 GHz, an electronic tuning range of 70 MHz, and a maximum output power of ~100 mW. The microwave frequency of one oscillator is measured using an EIP Model 575B frequency counter which has frequency locking capabilities. The frequency of the second oscillator can be varied independently and its frequency is set by monitoring the frequency difference of the two oscillators. Both channels are equipped with a 2-bit phase shifter (Vectronics DP629.5HS), that has a switching time of less than 50 ns, and allows 0°, 90°, 180°, and 270° phase shifts. The relative phase of the microwaves in the two channels can be further adjusted by manual phase shifters (ARRA, 6428-180XS for channel I and ARRA, 9424A for channel II), as shown in Fig. 1. Channel I and channel II are gated by a PIN diode microwave switch (M/A -COM ML6540-112D) with an ECL driver and a PIN diode microwave switch (General Microwave, DM864BH) with a TTL driver, respectively. Three mechanical MW switches select between four possible arrangements of the two channels (Fig. 1). Using these switches, it is possible to select a common MW source for channels I and II

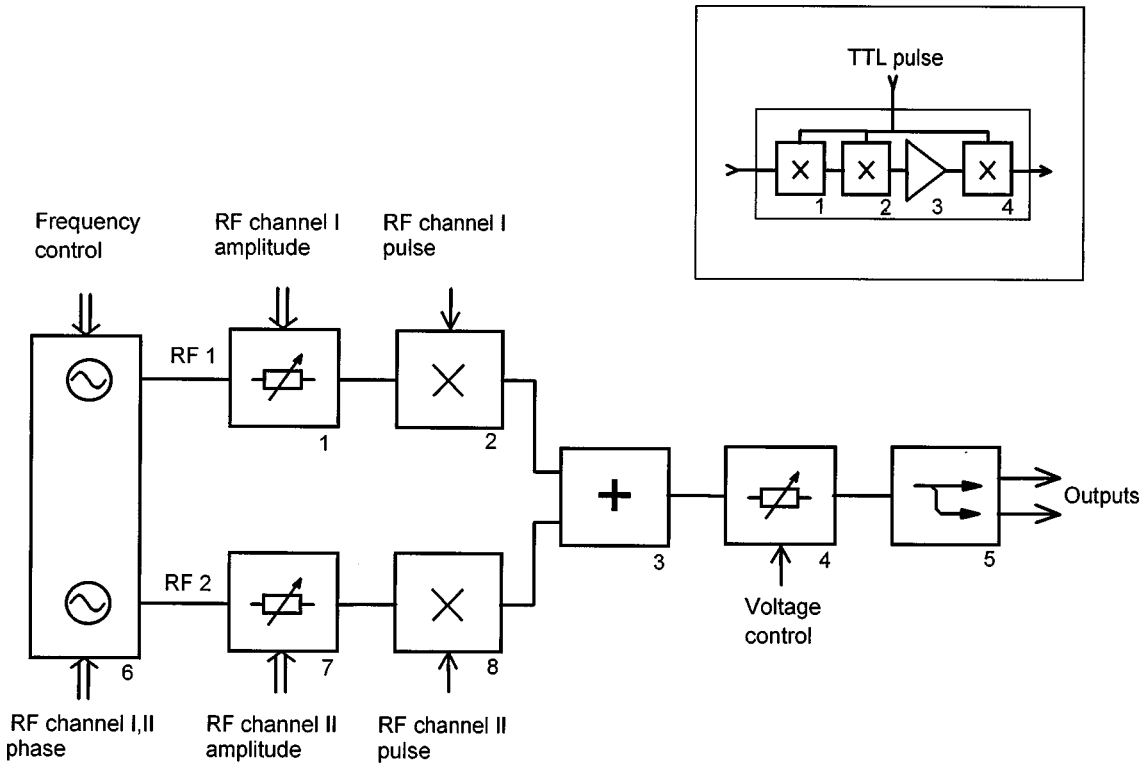


FIG. 2. Block diagram of the rf unit. 1,7–digital attenuators; 2,8–rf switches; 3–combiner; 4–voltage-controlled attenuator; 5–10 dB directional coupler; 6–dual channel radio frequency synthesizer. Inset: detailed structure of the rf switches (2,8): 1,2,4–double-balance mixers; 3–20 dB amplifier.

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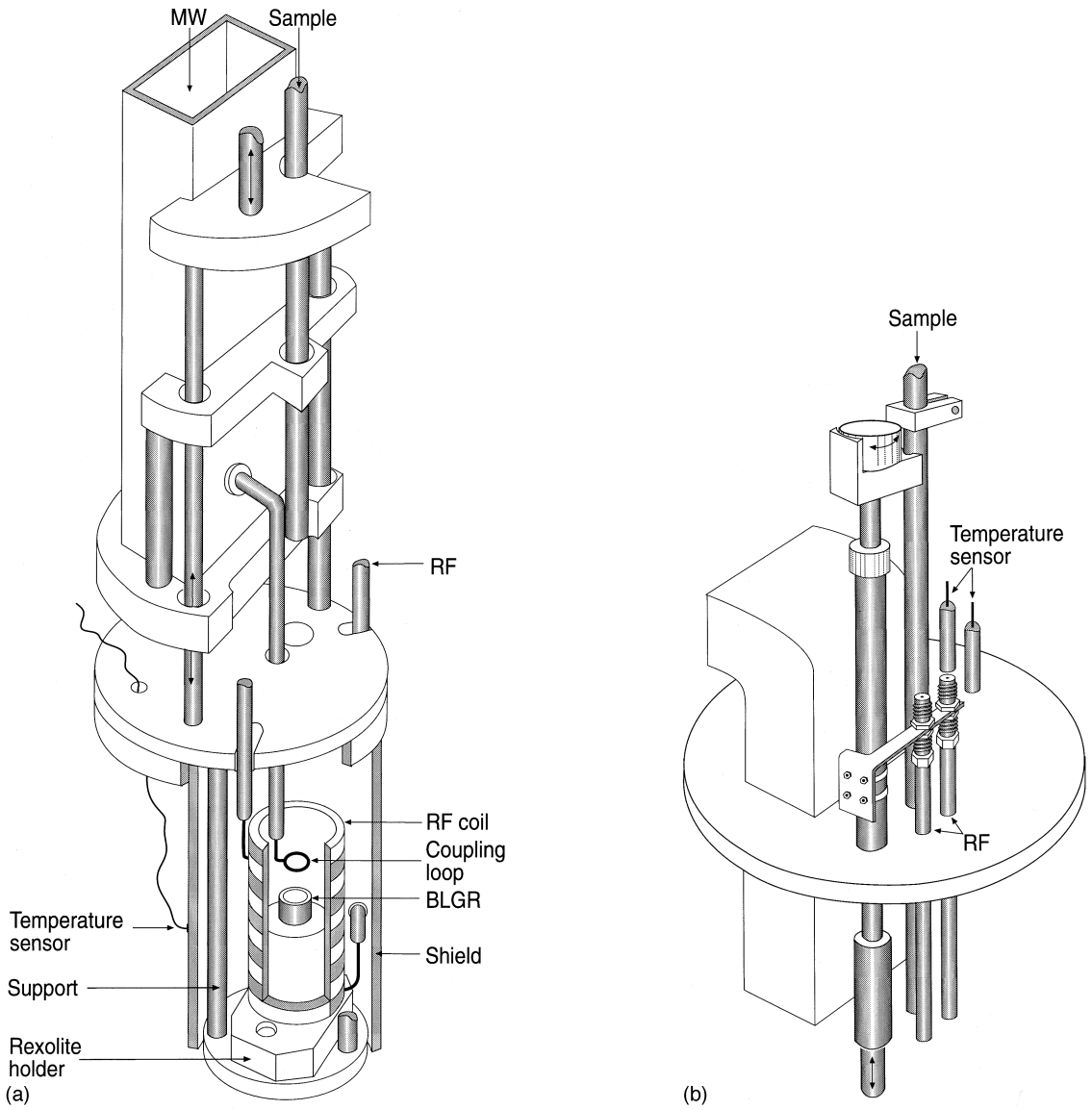


FIG. 3. The bottom (a) and top (b) parts of the pulse ENDOR probe head based on a bridged loop gap (BLG) resonator. The double sided arrows indicate the up/down movement of the resonator assembly that adjusts the MW coupling. This is achieved by rotating a knob at the top of probe head.

and to allow channel II to bypass the 1 kW traveling wave tube (TWT) amplifier (Applied System Engineering, Model 117X-1). This last configuration is used for generating low-power MW pulses such as required for the coherent Raman beat experiment.[20]

The microwave signals and the reflected high-power pulses are directed into the detection system, which consists of a limiter (Avantek), a protect PIN diode switch (General Microwave DM864BH), a high-pass filter, and a low-noise amplifier with a 1.5 dB noise figure at 36 dB gain (Miteq AMF-5S-080100-15). For electron-spin echo experiments, the amplified signal is mixed with a reference signal using a double balanced mixer (RHG, DBM8-12). In measurements requiring quadrature detection, such as FT-EPR experiments, a quadrature mixer (RHG, IRD10F23MC) is employed. The video signals from the mixer outputs are fed to broadband amplifiers (Penstock). This amplified signal is sampled by a boxcar integrator, in the case of echo detection, or by a digitizer (Nicolet 4094) for FT-EPR experiments and other experiments where the signal shape is important.

B. rf unit

The rf unit, shown in Fig. 2, is based on a dual channel PTS D310 synthesizer with a frequency range of 0.1–310 MHz. Two independent rf channels can supply pulses of variable length, frequency, amplitude, and phase (μs time scale). High speed phase switching to within 2.25° is achieved using the phase switching capabilities of the PTS D310. The rf amplitudes are controlled by high-speed, digital attenuators (Daico, DA-0897), providing a 0–63.5 dB attenuation range in 0.5 dB steps with a 60 ns switching time and a phase flatness tolerance of 3°. The rf pulses are produced using double-balance mixers, operating as rf switches, and 20 dB amplifiers (see inset in Fig. 2). The pulses from both channels are combined, adjusted by a voltage-controlled attenuator and fed into an amplifier (IFI 436P rf, 0.01–220 MHz) with a maximum output power of 3 kW in pulsed

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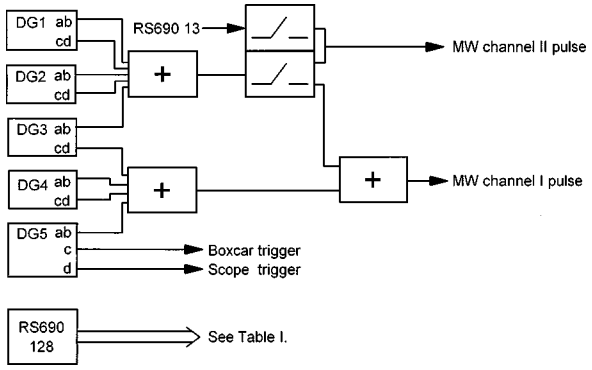


FIG. 4. A schematic representation of the pulse programmer, consisting of a digital word generator (RS690) and five DG535 delay generators (DG1-5), and its connections to the spectrometer. The boxes represent combiners (+) and switches (−/−), and the assignment of their various outputs.

mode, and 1 kW in continuous wave mode. The high power rf pulses are directly fed to the probe head. The output current of the rf pulses is measured using a 1 mV/1 mA Tektronix CT-2 rf current transformer.

C. Probe head

The ENDOR probe head is based on the design of the ETH laboratory.[21,22] It employs a bridged loop gap (BLG) resonator, coupled via a loop, and tuning is achieved by varying the distance between the loop and the resonator. The BLG resonator is made of silver-coated quartz where the silver paint used (Demetron, Leitsilber 200) does not require any burning or heating. The rf coil is a 13-turn, copper coil engraved on a Teflon tube that serves also as a MW radiation shield.[22] The probe head fits into a Janis Super Varitemp 7.75 DT cryostat which has a sample tube length of ~105 cm and a diameter of 30 mm. In the original design,[21,22] tuning of the coupling was achieved by an up and down motion of the coupling loop which was attached to a semirigid cable. Adaptation of this design to our probe head required a relatively long semirigid coaxial cable leading to mechanical problems during tuning that resulted in an off-centered loop position. Additionally, the relatively high loss of both pulse power and signal intensity over the long semirigid cable should be avoided. Both difficulties were overcome by replacing a major portion of the semirigid cable by a waveguide, and fixing the position of the loop while varying the position of the resonator holder.

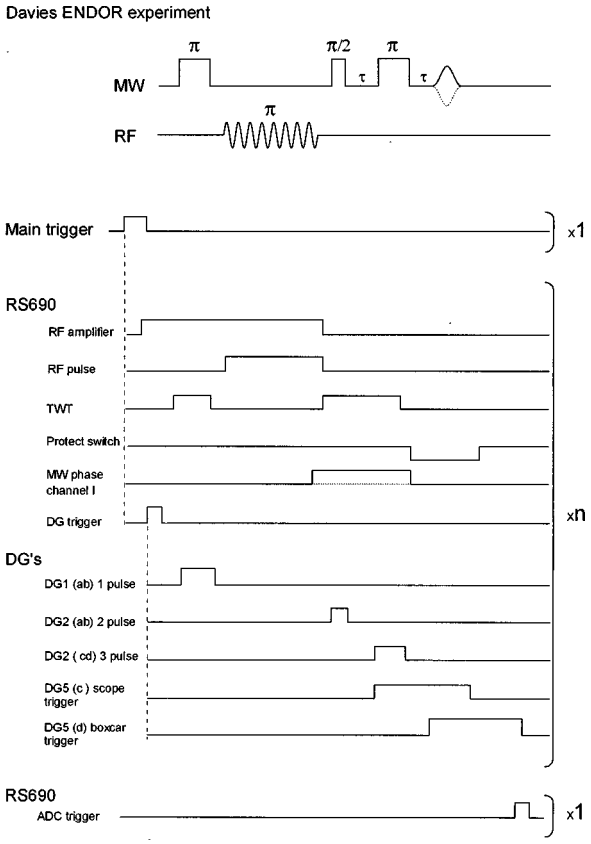


FIG. 5. Output pulse pattern of the pulse programmer for a Davies-ENDOR pulse sequence (shown at the top of the figure). The dotted and solid signals correspond to echoes generated when the three MW pulses have 0,0,0 or 0, π,π phases, respectively. The bottom part of the figure shows the corresponding timing diagram of the pulse programmer. The number of shots is represented by n.

A schematic presentation of the ENDOR probe head is shown in Fig. 3. The small diameter of the cryostat sample tube excluded the use of commercial waveguide/coaxial connector, necessitating a direct connection between the waveguide and a short semirigid cable with the coupling loop. The performance of the probe head is very sensitive to the quality of this connection. Consequently, great care should be exercised in its construction. The connection is centered on the wide section of the waveguide, at a quarter wavelength distance from the shorted end of the waveguide (WR-90). This point coincides with the maximum electric field position. The length of the central conductor of the coaxial line which

TABLE I. The assignment of the RS690 channels to the various devices.

Output channel	Device	Output channel	Device
1	TWT	11	DG trigger
2	Protect switch	12	ADC trigger
3	90° phase, MW channel I	13	MW pulse, channel II (optional)
4	180° phase, MW channel I	14–16	not used
5	90° phase, MW channel II	17–23	amplitude of rf, channel I
6	180° phase, MW channel II	24–32	phase, rf channel I
7	rf pulse, channel I	33–39	amplitude of rf, channel II
8	rf pulse, channel II	40–48	phase, rf channel II
9	rf amplifier	49–128	not used
10	rf control trigger		

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TABLE II. The PPL program of a Davies-ENDOR sequence, also designed to measure the nuclear nutation frequency. A phase cycle of $(0,0,0)$ $-(0,\pi,\pi)$ is applied to the MW pulses.

```
extern const time t=16 us
extern const time tau=410 ns
extern const time t90=30 ns
extern const time t180=60 ns
extern time trf=100 ns
program daviesrabi
trigger 1
mwpulse t180, channel=1, phase=[0, 0]
start begin
wait 200 ns
rfpulse trf
end
wait t
mwpulse t90, channel=1, phase=[0, 180]
wait tau
trigger 2
mwpulse t180, channel=1, phase=[0, 180]
wait tau
trigger 3
detect [1, -1]
```

TABLE III. The template file for a Davies-ENDOR sequence where in one dimension ([*x*]) the rf pulse width is varied and in the other dimension ([*y*]) the rf frequency. This experiment uses the pulse program given in Table II. Phase cycling is done in every scan rather than at every point, as determined by the ''order='' command.

```
[general]                          reprate=2 ms
template=TRUE                      ch2 on dg=false
description=Davies-Rabi ENDOR      [field]
detector=boxcar                    address=2048
band=X                             center=3300 Gauss
sample=CuA                         sweep=0 Gauss
files=1                            remote=false
[ sweep]                           gauss meter=false
shots=30                           [boxcar]
cycles=1                           sensitivity=0.5 V
size=100 31 1                      offset=0.0
order=x cycle scan y z             linear=true
[x]                                [rf]
trf=100 ns step 150 ns             rf1=20 MHz
[y]                                rf2=1 MHz
rf=1 MHz to 31 MHz                 mix mode=false false
[pulse]
program=c:\davrabi.ppl
```

is inside the waveguide was determined by maximizing the transmitted power reaching a diode detector located at the other end of the semirigid cable. The cable was then soldered to the waveguide, the diode was removed and the coupling loop was soldered to the cable.

The probe head is designed to allow frequent sample changes and houses two temperature sensors—one near the sample location and the other at a higher position for sensing the He level in the sample tube when pumping is required for temperatures below 4.2 K. The rf is fed through semirigid cables and two low-pass filters are inserted close to the rf coil to eliminate coupling to the microwave field.

D. Pulse programmer

The pulse programmer controls the MW and rf switches, phase settings, and the digital attenuators controlling the amplitudes of the rf pulses. It also provides trigger pulses to the oscilloscope, the boxcar integrator, and the digitizer. The pulse programmer (Fig. 4) consists of an Interface Technologies RS690, 250 MHz, digital word generator and five Stanford Research Systems DG535 digital delay generators. Auxiliary electronics were constructed to buffer the RS690 output signals, to combine the DG535 output pulses, and to trigger signal distribution. The RS690 and the DG535s are controlled by the computer via GPIB interfaces.

The RS690 has 32 independent output channels, when operating at its highest time resolution (4 ns). The number of channels can be increased to 128 when a time resolution of 16 ns is sufficient. Because of its limited resolution the RS690 is used for triggering devices which do not require subnanosecond accuracy, such as those employed for switching the MW phase shifters, the TWT amplifier gate signal, and for generating rf pulses (including the phases and amplitudes of the rf pulses). The 4 ns resolution is problematic when short MW pulses are desired, such as in FT-EPR experiments where wide excitation bandwidths are essential. This limitation is overcome by the use of the DG535s, which feature a very high timing precision (± 50 ps). This high accuracy is used for the operation of the MW switches which form the MW pulses. Each of the five DG535 can produce two independent pulses resulting in a maximum of ten pulses. The assignment of the various output channels of the RS690 and DG535s to the various devices is given in Fig. 4 and Table I. The DG535s are used to control up to nine MW pulses and to trigger the oscilloscope and the signal detector (boxcar integrator or digitizer). An example of a timing diagram for a Davies-ENDOR experiment[23] is shown in Fig. 5.

When the RS690 is used at a 4 ns time resolution, it can control the amplitude and phase of only one rf channel. Controlling both rf channels requires a reduction of the time resolution of the RS690 to 8 or 16 ns. This does not have a significant effect on the experiment, because the rf pulses are at least two orders of magnitude longer than these values and the MW pulses are not limited by the RS690 resolution.

To synchronize the RS690 word generator and the DG535s, one output of the RS690 word generator is used as a trigger pulse to the DG535s. This trigger pulse is buffered and distributed to the DG535s using cables of equal length to limit a channel-to-channel skew. Any residual skew can be corrected in the programming of the DG535s themselves. To reduce acquisition delay jitter and phase incoherencies between experiments, all pulse generators, and rf sources are phase locked to a common 10 MHz clock derived from the frequency counter. When signals are detected with the digitizer rather than the boxcar integrator, the common clock is derived from the digitizer.

E. Data acquisition program

The data acquisition program is written in C++ (Borland version 4.5) and uses the Windows-95 Multiple Document Interface (MDI) programming model. The program uses four parameter files which provide the information required for operation of the spectrometer. These are the ini-

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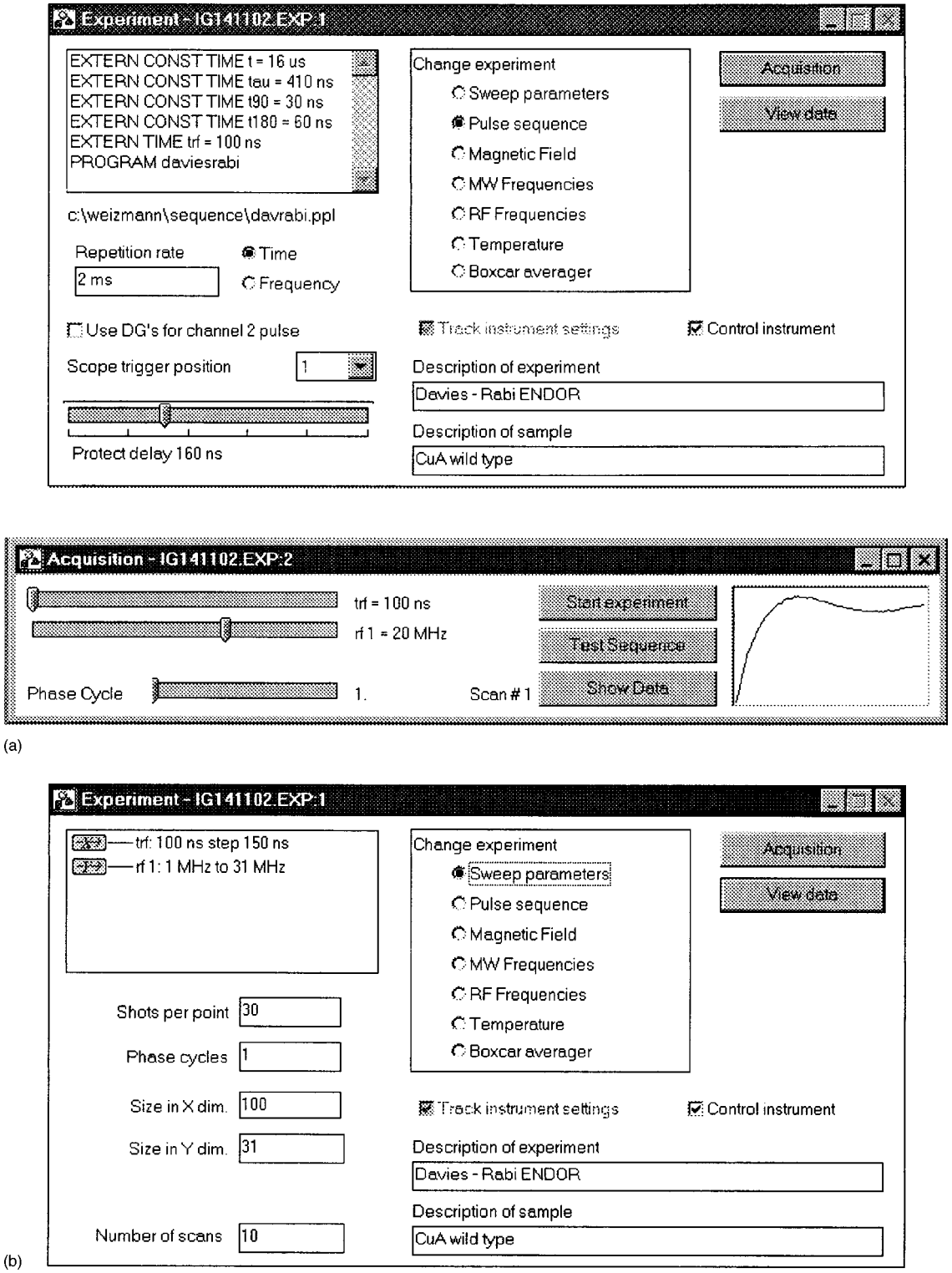


FIG. 6. (a) The user interface window of the data acquisition program displaying the experiment and the acquisition windows. In this figure the experiment window shows the pulse sequence parameters, (b) the experiment window showing the sweep parameters.

tialization, calibration, pulse program, and template files. The initialization and calibration files provide information concerning the hardware setup. They define the default settings of the spectrometer and specify values of calibration constants. The initialization file tells the program where to find the different devices on the GPIB bus and which inputs and outputs of the DAS card are used for various purposes. The calibration file assigns the output channels of the pulse programmer (RS690 and DG535s) to specific devices, and it also determines how the DG535 channels should be used (logic, impedance, etc.). The pulse program and template files contain all the necessary variables describing the experiment to be performed. The former defines pulse setup: the length and order of the pulses, the time interval between pulses, the phase cycling scheme, and the channels used. The template file contains details about the experiment: initial

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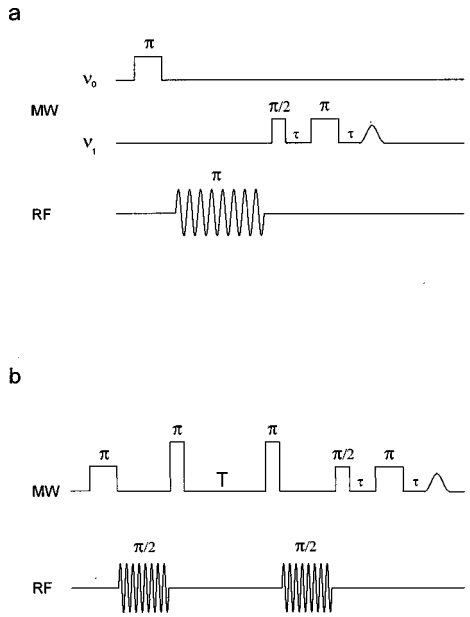


FIG. 7. Schematic representation of the HS-ENDOR pulse sequence (a) and the HYEND pulse sequence (b).

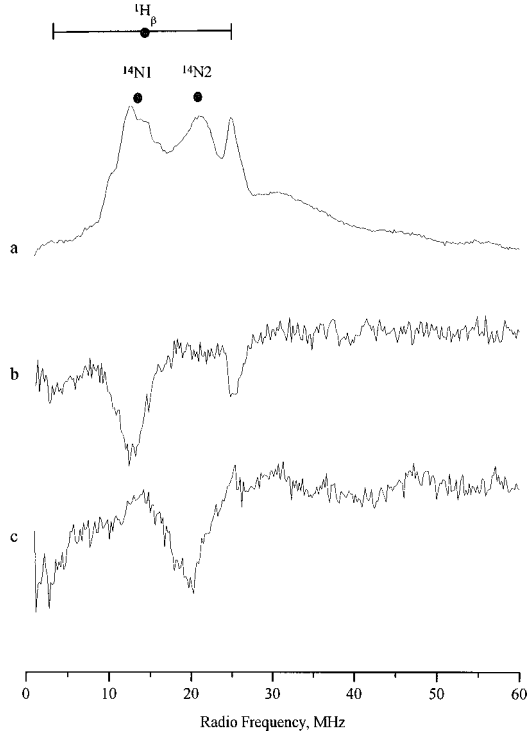


FIG. 8. HS-ENDOR spectra of a frozen solution of laccase obtained with Δ values of (a) 0 MHz; (b) 25 MHz; (c) 40 MHz. Other experimental parameters: $T=4.2$ K, $\nu_0=9.400$ GHz and $B_0=322.0$ mT. The length of the MW pulses were 40, 20, 40 ns in (a) and 160, 80, and 160 ns in (b),(c). The rf pulse length was 3.5 μs.

values and increments of the pulse program variables which are changed during the first (x), second (y), and third (z) dimensions of the experiment, the number of steps (points), the number of repeated measurements at each point (shots), the total number of scans, how the phase cycling is executed, and the repetition rate.

To allow an easy and user friendly control of the pulse programmer, a flexible, pulse programming language (PPL) was designed. As an example, the pulse program for a Davies-ENDOR sequence is shown in Table II. The first five lines are variable declarations and initialization. The variables *t*, and tau representing time intervals, and *t90* and *t180*, which are used to define the $\pi/2$ and π MW pulse lengths, are symbolic constants which can be set interactively, but do not change during the experiment. In contrast, the duration of the rf pulse, *trf*, is defined as a time variable with an initial value of 100 ns which is changed throughout the experiment. The command *trigger* provides the trigger pulse to the scope. Several *trigger* commands can be used in one program and the final selection is made later interactively. MW and rf pulses are applied using the *mwpulse* and *rfpulse* commands which also define the length, channel, and phase. The command *wait* defines a delay time while the *detect* command is used for the detector trigger and for the sign definition of the signal during a phase cycle. The phase cycling is generally indicated by an array of variables in square brackets (maximum length of 32) in the order of their execution. Table III lists the template file for a Davies-ENDOR experiment, which uses the *pulse program* given in Table II. It is designed as a 2D experiment where both the frequency and length of the rf pulse are varied. This measures the Rabi frequency and enables the optimization of the rf pulse length.

Most of the variables in the pulse program and in the template files can be manipulated interactively, using the Graphical User Interface of the acquisition program. Figures 6(a) and 6(b) show the User Interface windows of the pro-

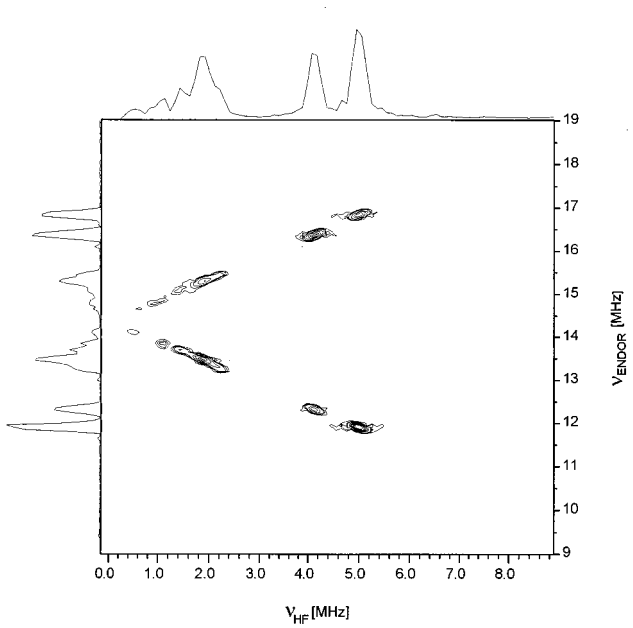


FIG. 9. Two-dimensional HYEND spectrum of a γ-irradiated single crystal of malonic (arbitrary orientation) recorded at 110 K. Experimental parameters: $\nu_0=9.472$ GHz, $B_0=334.7$ mT, the lengths of the MW pulse lengths were 200, 50, 50, 100, and 200 ns, and the rf pulse lengths were both 3 μs. The initial value for T was 100 ns, its increment 40 ns and 128 points were collected. The rf frequency increment was 50 kHz and 260 points were recorded. The phase cycle used for the rf pulses was $(0,0)-(0,\pi)-(\pi,0)+(\pi,\pi)$.

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gram for a Davies-ENDOR experiment, employing the pulse program and template files presented in Tables II and III, respectively. The experiment window shows all the fields (variables) which can be changed by the user, such as sweep parameters, pulse parameters, and magnetic field. An experiment can be run in *test sequence* mode or in *start experiment* mode [see Fig. 6(a), acquisition window]. In the former a single set of parameters of the multidimensional experiment, including the active phase cycle, is selected interactively by the user, employing the bars on the left-hand side of the window. This enables the optimization of signals and testing of the experimental setup. In the *start experiment* mode the experiment is run and its variables are automatically incremented by the computer, according to the parameters predefined in the template file.

F. Performance

1. Time resolution and speed

The initial trigger to the pulse programmer is given by the computer using the DAS1600 card and the repetition rate of a pulse sequence is controlled accurately by the RS690, thus avoiding computer interruption. The time required for loading the data to the RS690 and the DG535s should be minimized in order to keep the total duration of an experiment as short as possible. In our setup this time depends on the variables that are varied during the experiment. Variation of rf frequency in an ENDOR experiment is very fast ($\approx$1 ms/point), whereas changing magnetic field in a field sweep experiment requires 70 ms/point. The situation is worse when a time interval is varied. In a two-pulse ESEEM experiment employing both the RS690 and the DG535s, $\sim$150 ms/point is required to update the experimental parameters. This long ''communication dead time'' becomes most disturbing when 2D experiments, such as HYSCORE, are performed. For example, it adds a total of 41 min to a 128$\times$128 experiment for each phase cycle. Since the time required for the update of the DG535s is longer than that of the RS690 it is possible to reduce the loading time to $\sim$70 ms/point, at the expense of time resolution, by eliminating the DG535s. This is achieved by simple changes in the calibration file. In this setup the resolution will be 4 ns and the communication dead time reduces to 19 min/phase for a 128$\times$128 HYSCORE experiment.

2. Experimental results

Two examples are given below to demonstrate the performance of the spectrometer. The first is of an HS-ENDOR experiment,[8] [Fig. 7(a)], carried out on a sample of a frozen solution of the blue copper oxidase, laccase. The sample contains two types of EPR active Cu(II) centers, type I and type II.[24] The concentration of the protein was 1.4 mM in a 100 mM phosphate buffer, 50% glycerol (pH=6.0) solution and the sample tube used was a 3 mm o.d. quartz tube. The measurements were carried out at a magnetic field position near the $g_\perp$ of the two sites, and the experimental details are given in the figure caption. The Davies-ENDOR spectrum ($\Delta = \nu_0 - \nu_1 = 0$, Fig. 8), exhibits several overlapping signals of protons and nitrogens. The HS-ENDOR spectra obtained with $\Delta = 25$ MHz and $\Delta = 40$ MHz, are shown in Figs. 8(b) and 8(c). The broad peak at 20 MHz can be assigned to a nitrogen with a hyperfine coupling of 40 MHz; the peak at 12.5 MHz is attributed to another nitrogen with a coupling of 25 MHz, and the peak at 25 MHz is the high frequency component of a proton doublet with a coupling of 23 MHz.

The second example is of the HYEND (hyperfine-ENDOR) experiment[12] carried out on a γ-irradiated single crystal of malonic acid. The pulse sequence is shown in Fig. 7(b). This is a 2D experiment where both a time interval of the rf frequency are varied, and the phase cycle involves alternating the phases of the rf pulses. The spectrum obtained after Fourier transformation of the time domain dimension is shown in Fig. 9 and the experimental details are given in the figure caption. In this spectrum one axis gives the hyperfine coupling spectrum of the various protons while the other gives their ENDOR frequencies.

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